



Energy Industry

Trend Report





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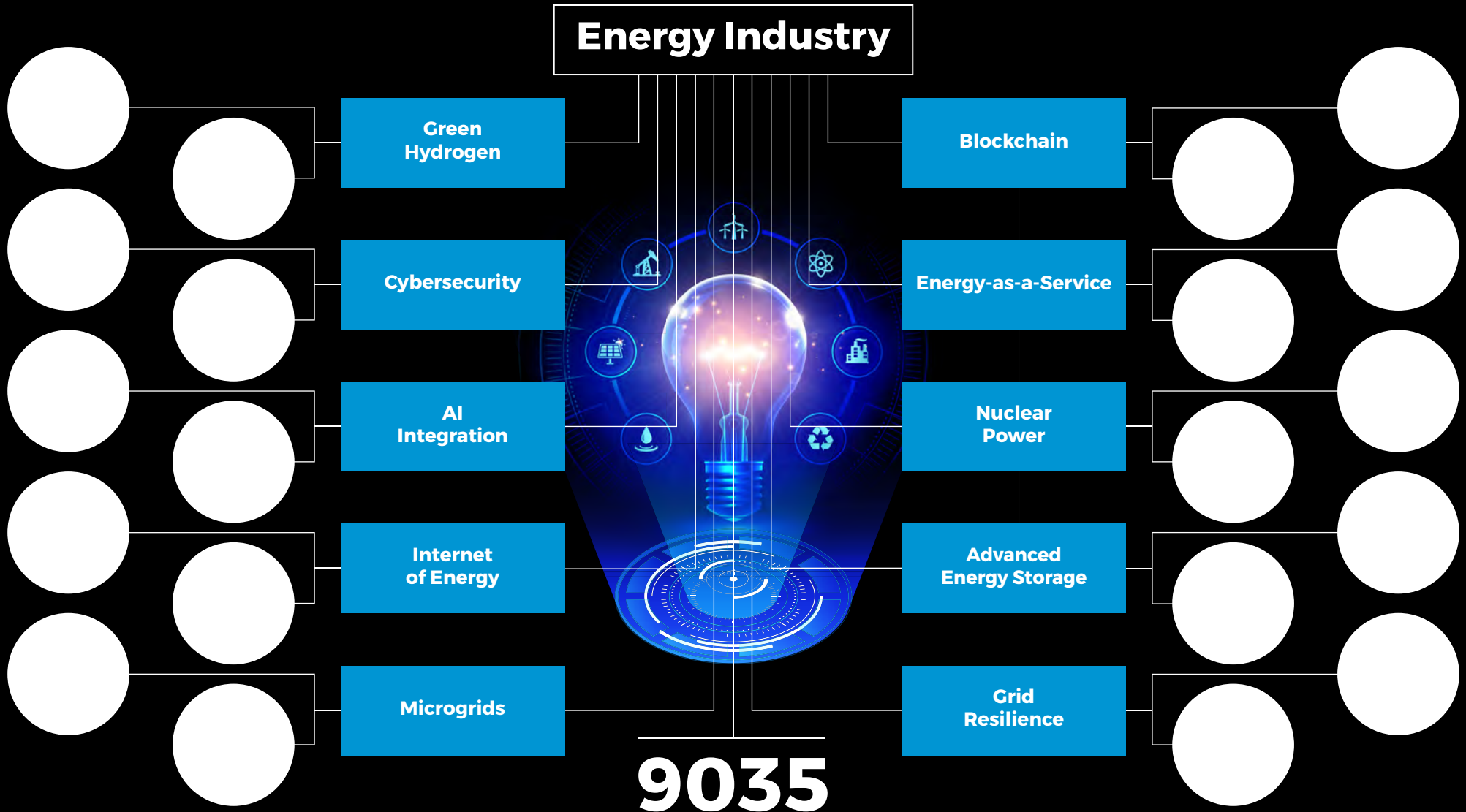
► The energy industry is embracing innovation to enhance efficiency, security, and sustainability. Green hydrogen, AI-powered optimization, advanced energy storage, microgrids, nuclear power, and grid resilience are some of the most critical trends impacting the energy sector.

► Covering over 7 million startups & scaleups globally, we use our [Big Data and AI-powered Discovery Platform](#) to identify innovative applications, technologies, and companies impacting the Energy Industry. This exhaustive, data-driven startup scouting pinpoints emerging trends and technologies in the Energy Industry. For this research, we analyzed a sample of 9035 startups and scaleups and present the Top 10 Trends along with 20 highly relevant solutions.

► In the Innovation Map below, you get an overview of the Top 10 Energy Industry Trends & Innovations that impact companies worldwide.



Innovation Map: Energy Industry



Startups & emerging companies analyzed

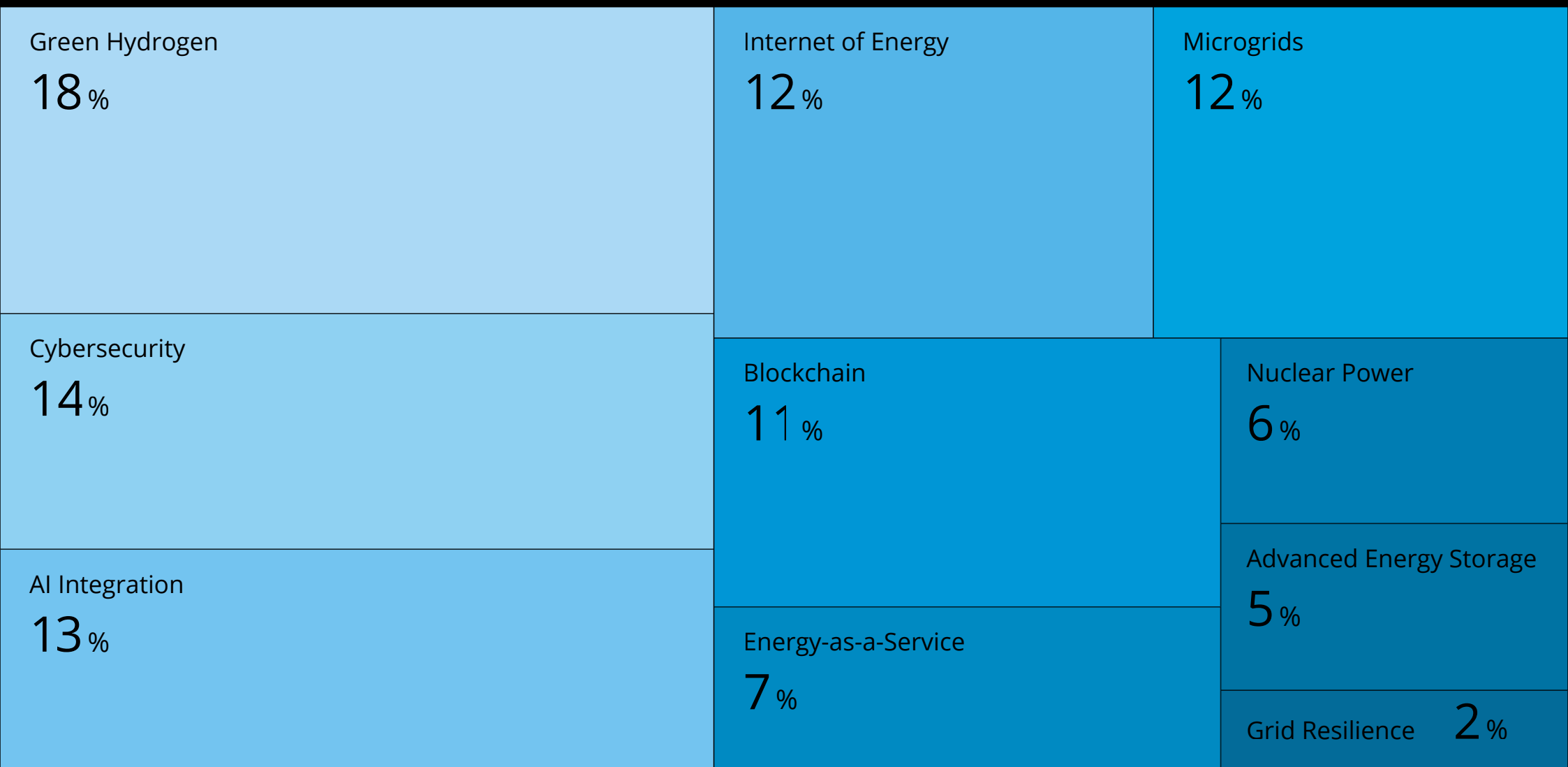


Tree Map reveals the Impact of the Top 10 Energy Industry Trends

► The Tree Map below illustrates the major energy developments. Growing investments in microgrids and grid resilience support sustainability, which is also driven by green hydrogen and advanced energy storage.

► Blockchain, IoT, and AI integration improve security and efficiency, while energy-as-a-service and nuclear power transform how energy is produced and used.

Top 10 Energy Industry Trends & Innovations





Global Startup Heat Map covers 9035 Energy Startups & Scaleups

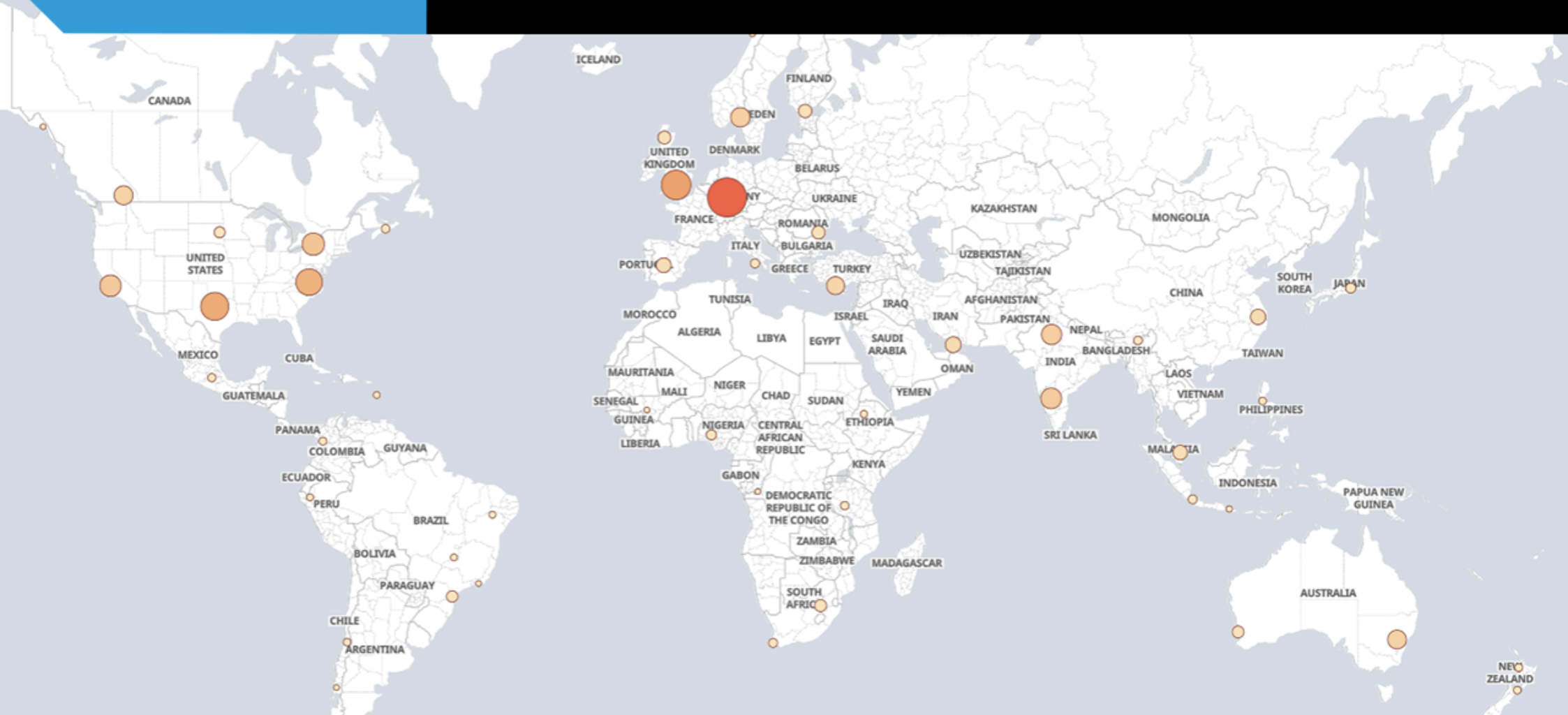
► The Global Startup Heat Map showcases the distribution of 9035 exemplary startups and scaleups analyzed using the StartUs Insights Discovery Platform. It highlights high startup activity in Germany and France, followed by the US & the UK.

► From these, 20 promising energy startups are featured below, selected based on factors like founding year, location, and funding.

9035

STARTUPS ANALYZED

Global Startup Heat Map: Energy



This Global Startup Heat Map illustrates the geographical distribution of 9035 startups & emerging companies we analyzed for this topic

Green Hydrogen

▶ Green hydrogen is rapidly emerging as a crucial element in the energy industry's transition towards sustainability. Since green hydrogen is created from renewable energy sources like solar and wind, it produces no carbon emissions – unlike gray and blue hydrogen. In industries where electrification alone is inadequate, like power generation, heavy industries, and transportation, green hydrogen is becoming more popular.

▶ Norwegian startup [Element One Energy](#) develops high-efficiency electrolyzers and fuel cell systems to accelerate the adoption of green hydrogen. This technology produces hydrogen at high pressure without the need for extra compression by utilizing proton exchange membrane (PEM) electrolysis. This boosts energy efficiency and cuts expenses. The technology is integrable into energy storage, mobility, and industrial applications due to its small size. It reduces the need for gas treatment and yields high-purity hydrogen as well as high-pressure oxygen as a valuable by-product. The startup's fuel cell systems support clean power generation, which converts hydrogen into electricity. It enables a shift toward carbon-free energy and industrial decarbonization. Element One

Energy makes hydrogen a feasible energy carrier by developing affordable and scalable hydrogen solutions.

▶ Norwegian startup [SOREN HYDROGEN](#) creates a strong digital planning tool to expedite the net-zero transition and green hydrogen projects. The platform expedites project development and enables stakeholders to make well-informed decisions quickly through the integration of financial modeling, resource evaluation, and data analytics. The platform integrates various components — such as wind, solar, power purchase agreements (PPA), grid, electrolyzers, balance of plant (BoP), and storage sizing. This optimizes levelized cost of hydrogen (LCOH) using top-tier proton exchange membrane (PEM) and alkaline water electrolysis (AWE) electrolyzers. The tool's intuitive interface makes scenario analysis easier and optimizes project parameters to increase cost-effectiveness and energy efficiency. It enables enterprises to adopt green hydrogen solutions by offering thorough insights into renewable hydrogen production. SOREN HYDROGEN's digital planning tool accelerates the adoption of green hydrogen with the help of electrification, increased energy efficiency, the deployment of renewable energy sources, and full decarbonization.

Cybersecurity

▶ The energy sector is more vulnerable to cyberthreats such as ransomware, data breaches, and operational disruptions due to legacy systems and growing digitization. The energy industry uses cloud security, blockchain-powered transactions, AI-driven threat detection, and zero-trust architecture (ZTA) to reduce risks. These technologies allow for quicker decision-making while increasing efficiency, safety, cost effectiveness, and transparency.

▶ Australian startup [Cell Cyber](#) develops advanced cybersecurity technologies to aid organizations mitigate cyber risks and strengthen digital resilience. It combines predictive cybersecurity, real-time threat monitoring, and customized security training to proactively combat changing cyberthreats. Its AI-powered threat intelligence solution enables companies to react quickly to cyberattacks by detecting vulnerabilities before exploitation. The startup improves defenses against advanced cyberattacks, ransomware, and phishing by incorporating automated security monitoring, penetration testing, and ongoing risk assessment. Its security training programs provide organizations with the necessary skills to recognize, stop, and react to threats to assure a human-centric defensive strategy. Cell Cyber enables businesses to protect vital assets,

uphold compliance, and construct a strong digital infrastructure by emphasizing proactive protection, automation, and adaptability.

▶ US-based startup [Cyware](#) creates cyber fusion solutions that assist enterprises in advancing security operations by combining automated threat response, collaboration, and threat intelligence sharing. Its platform enables security professionals to connect different security data, proactively block threats, and speed up response times, which lessens the burden of repetitive activities. This allows businesses, government organizations, managed security service providers (MSSPs), and information sharing and analysis center (ISAC)/ISAOs communities to work together more effectively in real time by simplifying the intake, correlation, and distribution of threat intelligence. Cyware's automated threat response and orchestration allows organizations to reduce the amount of manual intervention required to contain security incidents which enhances productivity, decision-making, and overall resilience. The startup improves security visibility and control while fortifying defenses against changing cyberthreats. By automating reaction activities and operationalizing threat intelligence.

AI Integration

▶ The energy sector faces challenges like demand fluctuations, inefficient grids, and manual labor. AI integration optimizes energy distribution and consumption by addressing these problems. AI-driven autonomous energy management systems (EMS), smart grids, and digital twins enhance efficiency, sustainability, and cost-effectiveness. These technologies enable real-time monitoring, predictive maintenance, and automated energy trading to reduce downtime and resource wastage. Additionally, digital twins create virtual replicas of energy infrastructure for optimizing performance and ensuring grid reliability.

▶ French startup [DEAP VISION](#) develops AI-powered Health, Safety, and Environment (HSE) software to automate industrial workflows and enhance operational safety. The AI-HSE platform ensures adherence to international safety standards by utilizing digital technology and AI to identify safety infractions, send out real-time notifications, and predict possible risks. It allows organizations to monitor environmental impact, manage regulatory compliance, and optimize safety practices by integrating modern data analytics. The cloud-based platform centralizes HSE data and provides risk assessment tools and predictive insights to minimize wasted time, decrease risks,

and enhance decision-making. It enables organizations to increase operational efficiency and improve worker safety by reducing human error and expediting incident reporting. DEAP VISION's automated solutions aid businesses in increasing productivity and revenue through astute, data-driven safety management by lowering liabilities and guaranteeing workplace sustainability.

▶ Indian startup [Bharat Flow Analytics](#) offers advanced pipeline monitoring and inspection solutions for the energy sector, utilizing non-intrusive technologies to enhance pipeline integrity. Its solutions, iPTran and iLDS, employ enhanced pressure transient technology integrated with AI and the industrial internet of things (IIoT) to deliver real-time insights into pipeline conditions. These devices determine the amount of wax and debris buildup, identify leaks quickly, and evaluate the general condition of the pipeline. Bharat Flow Analytics increases productivity while cutting down on maintenance expenses and downtime compared to conventional intrusive monitoring techniques. The system aids energy firms to preserve the integrity of pipeline infrastructure through obstruction prevention, flow rate optimization, and adherence to safety and environmental laws.

Internet of Energy

▶ The energy sector faces challenges from outdated infrastructure, unpredictable energy demand, and grid inefficiencies. The internet of energy leverages IoT sensors, smart meters, and cloud-based analytics to facilitate automated energy management, predictive maintenance, and real-time monitoring. These technologies optimize energy distribution and minimize waste, which, in turn, increases efficiency, transparency, and cost savings. IoE ensures safe, data-driven energy automation and trading through the integration of blockchain and AI, resulting in a strong, intelligent, and flexible energy ecosystem that facilitates the global switch to renewable energy sources.

▶ UK-based startup [ivBOT](#) provides an integrated platform for managing and optimizing renewable energy assets such as solar, wind, green hydrogen, and battery storage. Its Central Monitoring System (CMS) compiles asset performance data into a single dashboard while providing real-time analytics, insights into predictive maintenance, and tools for efficiency improvement. The platform's AI-driven algorithms minimize energy losses and increase production by identifying inefficiencies, anticipating maintenance requirements, and improving overall system efficiency. ivBOT reduces downtime and simplifies operations for energy companies by offering

automated control and remote access. The technology enhances load balancing and grid stability by enabling smooth coordination between several renewable assets. The startup provides companies more insight into asset performance, which aids them in meeting sustainability targets and cutting expenses.

▶ South Korean startup [CONPORTLAB](#) offers PORTA IoT EV Charger Link, a comprehensive EV charger management solution. It provides real-time monitoring, predictive maintenance, and energy optimization for electric vehicle charging infrastructure. The platform interfaces with EV chargers to provide automatic diagnostics, usage statistics, and intelligent load balancing to improve charging efficiency and lower energy costs. It facilitates user administration, automated problem detection, and remote monitoring of charging stations by utilizing cloud-based analytics for fleet operators and charging network providers. The system optimizes power distribution and schedules smart charging sessions, improving grid stability by maximizing uptime and performance reliability. PORTA IoT EV Charger Link's integrated access control and automated payment features improve user experience and expedite processes.

Microgrids

▶ Microgrids provide localized, independent energy solutions that improve resilience by reducing dependence on centralized electricity. Powered by renewable energy, battery storage, and intelligent energy management systems, microgrids support secure energy transactions, predictive maintenance, and real-time monitoring using blockchain, AI, and more. Microgrids ensure a steady supply of electricity and improve decision-making by lowering costs, increasing transparency, and optimizing energy distribution. Microgrids improve safety, dependability, and customer satisfaction by managing heavy loads with little human involvement.

▶ US-based startup [Direct Energy Partners](#) develops DCIDE, an advanced microgrid design software that simplifies the engineering and deployment of ultra-efficient decentralized microgrids. It offers a quick microgrid design framework, and automated design rule verification to ensure component compatibility and adherence to DC microgrid standards. DCIDE significantly lowers the soft costs of microgrid engineering, planning, and procurement by simplifying the design process. The platform incorporates real-time modeling tools to improve grid resilience, optimize electricity distribution, and assess system performance. Its intelligent configuration

capabilities enable companies to create sustainable, scalable energy networks that optimize the use of renewable energy sources while reducing system losses. Direct Energy Partners' products ensure long-term energy stability, cost effectiveness, and sustainability for industrial and commercial applications by promoting the deployment of decentralized microgrids.

▶ Chinese startup [Elecod](#) creates integrated microgrid multi-energy systems to maximize the distribution of energy in different industries. Its technology combines energy storage devices, sophisticated control algorithms, and renewable energy sources to provide strong and effective microgrids. The system's smooth integration of solar, wind, and battery ensures storage power delivery, which balances energy supply and demand. The platform's predictive analytics and real-time monitoring improve operational effectiveness and lessen dependency on conventional power grids. Elecod's modular architecture aids communities and businesses to implement scalable microgrids that meet their energy requirements. Its technologies offer affordable and environmentally friendly energy substitutes by supporting demand-side management, peak shaving, and grid stability.

Blockchain

► The energy industry faces challenges like lack of transparency, inefficiencies in energy trading, and grid security risks. Blockchain technology solves these problems through smart contracts, real-time traceability, and safe, peer-to-peer (P2P) energy transfers.

By integrating tokenized carbon credits, peer-to-peer energy trading, and tamper-proof records, blockchain boosts sustainability, lowers costs, and increases efficiency.

Direct trading between energy providers and customers is made possible by its optimization of grid management, fraud reduction, and data quality. Blockchain speeds up procedures, improves security, and creates a transparent, effective, and resilient energy network with automated, trustless transactions while removing intermediaries.

► Swiss startup [C4E](#) provides blockchain technology for the energy sector, enabling safe and open exchanges between grid operators, producers, and consumers. Its decentralized ledger technology facilitates peer-to-peer energy trade, automates contract execution through smart contracts, and ensures data integrity. The platform expedites compliance and verification procedures by facilitating grid flexibility, carbon

credit tracking, and energy certificate digitization. C4E improves energy transaction visibility and trust by combining blockchain technology with smart metering and IoT devices. The system facilitates decentralized renewable energy exchanges, minimizes inefficiencies, and optimizes trading procedures for energy market participants. Its flexible framework supports a wide range of energy assets, from small-scale solar generators to large utilities, promoting an open and just energy economy.

► UAE based startup [Solareum](#) develops a Layer 1 blockchain platform dedicated to green energy initiatives, providing a secure and scalable foundation for tokenizing renewable energy assets. Its blockchain platform allows businesses and individuals to invest in renewable energy while maintaining responsibility due to the transparent, decentralized trade of energy credits. The platform incorporates smart contracts by automating energy transfers and removing middlemen, which lowers administrative expenses. Solareum's ecosystem facilitates the tracking and verification of renewable energy generation with real-time transparency into sustainable energy projects.

Energy-as-a-Service

▶ The energy industry faces challenges like high upfront costs, inefficient energy management, and complex infrastructure maintenance. Energy-as-a-service tackles these issues by improving demand response and smart grid optimization as well as delivering subscription-based energy solutions. EaaS providers use IoT, cloud analytics, and predictive maintenance to maximize energy use, minimize waste, and cut expenses. This strategy increases efficiency and transparency for businesses through flexible, pay-as-you-use energy solutions.

▶ Swiss startup [nextE](#) provides advanced energy generation solutions to enhance efficiency, reliability, and sustainability across diverse applications. It offers solar, wind, and hybrid power systems for commercial, residential, and industrial applications. NextE minimizes environmental effects and maximizes energy output through the use of advanced engineering and materials. The startup's scalable modular energy solutions allow consumers to increase their generation capacity as demand increases. Its predictive maintenance and real-time monitoring features maximize system performance by minimizing downtime and guaranteeing long-term efficiency. The startup enables communication with distributed

energy networks in an uninterrupted manner, improving grid stability by incorporating smart grid connectivity. Its focus on sustainability aids with worldwide decarbonization initiatives by offering affordable, renewable energy options.

▶ Indonesian startup [360energy](#) allows customers to use renewable energy solutions without incurring large upfront expenses by providing a pay-as-you-go (PAYGO) subscription model for energy access. Its platform combines data analytics, mobile payments, and smart metering to offer a range of flexible energy consumption options. The technology improves the accessibility and cost of sustainable energy by enabling users to pay for energy according to usage. The PAYGO concept is beneficial in areas where traditional electricity availability is restricted in underserved communities. 360energy's technology ensures a smooth customer experience by enabling automatic invoicing and remote monitoring of energy consumption. The system provides communities access to sustainable energy options by facilitating the installation of mini-grids, solar household systems, and battery storage devices. It improves demand forecasting and energy efficiency by the platform's integration of IoT connections.

Nuclear Power

▶ Growing electricity demand, carbon emissions, and grid dependability issues are critical challenges for the energy sector. In response, advancements in fusion technology, small modular reactors (SMRs), and molten salt reactors are enhancing the safety, scalability, and economic viability of nuclear power, reinforcing its role as a reliable, low-carbon energy source. When combined with renewable energy, nuclear fusion, SMRs, and molten salt reactors increase sustainability, lower operational risks, and boost efficiency for a balanced grid. Optimizing reactor performance with AI-driven monitoring, automation, and predictive maintenance further increases safety, lowers costs, and ensures transparency.

▶ French startup [Calogena](#) creates sustainable heating solutions by employing small modular reactor (SMR) technology to create nuclear heat generators with zero carbon emissions. Its 30 MW thermal water-cooled SMR is designed for district heating applications, offering a compact and efficient alternative to fossil fuel-based systems. The reactor's low-pressure, low-temperature operation improves safety while reducing building timeframes and enabling factory assembly. Calogena ensures dependable heat generation with minimal environmental effects by utilizing established nuclear technology.

The system includes advanced passive safety safeguards to eliminate the risk of catastrophic failure while maintaining continuous operation. The startup's nuclear heat generator aids achieve worldwide decarbonization objectives by lowering CO2 emissions related to heat production.

▶ Italian startup [CAELUS](#) creates software solutions to expedite nuclear reactor licensing procedures and ensure compliance with changing nuclear safety regulations. Its digital platform lowers administrative workloads and speeds up approval processes by automating documentation, regulatory filings, and compliance tracking. CAELUS reduces human error and improves regulatory file accuracy by combining workflow automation and AI-driven data validation. The system facilitates cooperation amongst stakeholders, including regulators, plant operators, and engineering teams, and ensures conformity with international nuclear safety laws. The startup offers real-time license status tracking, which enhances decision-making and transparency in nuclear energy projects. Its technology streamlines nuclear licensing processes, lowers costs, and makes it easier to install next-generation nuclear reactors.

Advanced Energy Storage

▶ Advanced energy storage technologies reduce peak demand costs, renewable energy intermittency, and system instability. Solutions like solid-state, lithium-ion, and zinc-air batteries, as well as pumped hydro and hydrogen storage, address these problems. Solutions like solid-state, lithium-ion, and zinc-air batteries enable real-time energy management, load balancing, and demand response. This ensures increased sustainability and efficiency. Advanced storage also provides scalable and affordable solutions for households and businesses to support a resilient, low-carbon future.

▶ Turkish startup [Advenst](#) creates zinc-air energy storage technologies to give renewable power systems high-capacity, cost-effective energy storage. Its zinc-air batteries, which use abundant and recyclable ingredients, provide a safer and more environmentally friendly substitute for conventional lithium-ion storage. The technology is suitable for off-grid applications, grid balancing, and renewable energy integration as it allows for long-duration energy storage. Advenst's technologies provide long-term operating reliability by providing high energy density with little deterioration. The startup's scalable modular design enables utilities and companies

to implement adaptable storage systems that meet changing energy needs. The startup encourages ecologically friendly energy storage and improves supply chain resilience by lowering reliance on essential raw materials. This zinc-air technology strengthens energy security and facilitates the shift to sustainable energy storage by allowing for more effective utilization of renewable resources.

▶ South African startup [Hubble Energy](#) creates lithium-ion battery technology for scalable energy storage options for commercial, industrial, and household uses. It offers containerized solutions and high-voltage racks for large-scale energy storage requirements. The high-voltage racks provide effective energy storage with quick deployment capacity while the container solutions offer flexible, transportable storage choices that easily incorporate into current power infrastructures. Hubble Energy's Cloudlink platform systems offer smart management and real-time monitoring, allowing customers to maximize efficiency and ensure system dependability. The startup promotes the switch to renewable energy sources and increases energy resilience by providing adaptable and effective energy storage solutions.

Grid Resilience

► Extreme weather events, cyber attacks, and legacy infrastructure cause grid instability and disruptions. Energy storage, microgrids, AI-driven predictive maintenance, and self-healing grid technologies identify such problems, reroute electricity, and automatically restore services. This contributes to increased grid resilience. Such developments increase energy distribution's transparency, efficiency, and downtime. Utilities are also able to manage heavy workloads with lesser assistance from humans with automation and real-time monitoring. This results in lower costs, increased safety, and enhanced decision-making.

► US-based startup [NAVIA ENERGY](#) creates AI-powered digital twin platforms to improve grid stability and energy resilience. Its technology enables real-time monitoring and predictive analysis by building detailed virtual models of microgrids and distributed energy resources (DERs). The platform predicts system behavior and optimizes reactions by simulating a variety of scenarios, such as unfavorable weather events, cyberattacks, and grid disruptions to provide a continuous power supply. The solution aids operators achieve net-zero

targets and efficiently manage energy budgets with its mission-oriented decision support and forward-looking planning capabilities. It also offers intelligence-driven energy dispatch and cyber threat detection to ensure operational security. It supports services like reactive power management, demand response, and grid stability, which improves power delivery quality and the environment. The startup enables users to attain cost savings, environmental sustainability, and energy security utilizing AI and digital twin technologies.

► UK-based startup [Solus Power](#) specializes in off-grid rapid electric vehicle (EV) charging and energy storage solutions, addressing the growing demand for flexible and sustainable energy infrastructure. Its product Titan is an energy storage and off-grid, fast EV charging device for both commercial and military use. Titan is beneficial for a range of settings, such as parking lots at airports, car-rental agencies, and isolated or disaster-affected locations due to its portability and quick deployment. It offers effective charging solutions that improve operational efficiency as it operates independently of conventional power networks.



Discover all Energy Trends, Technologies & Startups

► The energy sector is set for major growth, with smarter grids, better energy storage, and decentralized power becoming more common. AI-powered energy management, advanced batteries, and hydrogen solutions will make energy systems more efficient and reliable. As the energy resilience market nearly doubles by 2032, investments in smart grids, microgrids, and renewables will help create a more secure and sustainable future.

► The Energy Trends & Startups outlined in this report only scratch the surface of trends that we identified during our data-driven innovation & startup scouting process. Identifying new opportunities & emerging technologies to implement into your business goes a long way in gaining a competitive advantage.



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